Game Systems

> *"**Game loop**: every frame, `GameScreen.render()` runs: events (sleep, cutscene) → player input → `enemies.update()` → `world.step()` → remove destroyed bullets and dead enemies → door check → camera → drawing.*
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> ***Physics**: Box2D world with no gravity, `PPM = 16` (16 pixels = 1 meter). Walls come from the Tiled "collisions" layer. Each map has its own World.*
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> ***Combat**: bullets are Box2D sensors. `GameContactListener` checks `BulletOwner`: a PLAYER bullet only damages enemies, an ENEMY bullet only damages the player (no friendly fire). `takeDamage()` lives in `Entity`, so it is shared by the player and the enemies.*
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> ***Weapons**: ammo and reload are handled in `Weapon`. Clicking with an empty magazine starts a reload; the gun can't shoot while reloading. `WeaponController` swaps between pistol and rifle and plays the shot and reload sounds and animations.*
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> ***Enemy AI**: `EnemiesManager` stores every enemy as an `Enemy` and calls `update()` on each one (polymorphism). Zombie: walks to the player, hits every 2 s while touching him. Military: keeps a random distance and shoots, or fights in melee without a weapon.*
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> ***Death**: health reaches 0 → the enemy stops and loses its collisions; its corpse stays on the floor and it is removed from the logic after 5 s.*
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> ***Maps & events**: Tiled "events" layer: named spawn points, doors (`target`, `spawn`, optional `cutscene`), bed. Changing map reloads the World, the lights and the managers; the weapon and its ammo are kept.*
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> ***Lighting**: box2dlights, walls cast shadows. Lamps, cones, sun and TV come from the Tiled "lights" layer. Sleeping in the bed switches between day and night. Each shot makes a muzzle flash.*
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> ***Post-processing**: the frame is drawn into a FrameBuffer, then a shader adds warmth, lower saturation, vignette, film grain and chromatic aberration.*
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> ***Audio**: positional sounds; volume depends on the distance (heard up to 400 px), panned left or right, with a small random pitch change."*